MIL-M-18221C(YD)
22 August 1983
SUPERSEDING
MIL-M-18221B(YD)
14 May 1974

MILITARY SPECIFICATION

METERS, FLOW RATE INDICATING

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flow meters that indicate, record, and totalize the fluid properties of air, gas, steam, and water. The meter is actuated by a primary sensing device that responds to the movement of a continuous fluid stream. A secondary element directly or indirectly translates the flow to the indicator or recorder.

1.2 Classification. Flow meters shall be of the following types and styles, as specified (see 6.2).

Type I - Differential pressure

Style A - Venturi
Style B - Diaphragm, or rotary
Style C - Orifice plate

Type II - Velocity

Style D - Propeller or vane
Style E - Turbine

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6680

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MILITARY

MIL-T-152 - Treatment, Moisture-and Fungus-Resistant, Of Communications, Electronic, And Associated Electrical Equipment.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129 - Marking For Shipment And Storage.
MIL-STD-794 - Parts And Equipment, Procedures For Packaging Of.
MIL-STD-1188 - Commercial Packaging Of Supplies And Equipment.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

390.03 - Gear Handbook - Volume 1 - Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1330 Massachusetts Avenue, N.W., Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B31.1 - Power Piping.
C39.4 - Automatic Null-Balancing Electrical Measuring Instruments, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

PTC 19.5 - Application, Part II of Fluid Meters: Interim Supplement on Instruments and Apparatus.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The meter shall consist of a primary sensing element and a secondary indicating element or combined indicating-recording-totalizing unit, which shall measure the system properties (see 6.2). Primary sensing elements, listed in table I, shall be composed of materials that will not chemically react with the air, gas, steam, or water to be metered. The secondary element shall receive the sensing signal from the primary element. The meter shall not be adversely affected by vibration and shall be easy to maintain.

TABLE I. Meter performance data.

Sensing Element	Type Meter	Differential pressure range* inches of water	Operating pressure * Maximum Pounds per square inch	Operating temperature range* °F	System accuracy percent full scale flow	Operating fluid or medium
Venturi	I	0-1000	700	0-800	± 1.5	steam, air, gas water
Diapragm, or rotary	I	0-1000	700	0-800	± 1.5	steam, air, gas water
Oriface plate	I	0-1000	700	0-800	± 1.5	steam, air, gas water
Propeller or vane	II	Capacity range. gallons per minute. 90-16,000	3,000	0-150	± 2	Liquids
Turbine	II	1/4-1250	5,000	-300 to +500	± 1.5	Liquids air, gas

Meter types are capable of operation under conditions within the limits indicated. Operation, for purposes of this specification, is not necessarily intended to extend over the entire range indicated. Typical requirements for metering applications within the above limits are noted in section 6. Actual range of operating conditions shall be as specified in 6.2.

3.2 Standard commercial product. The meters shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the meters being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.2.1 Identification or model numbers. The standard commercial product provided under this specification shall be identifiable by the manufacturer's regular, or by commercial, parts distribution or service organizations.

3.3 Design. The meter design shall permit ease of installation, accessibility for maintenance and service, and shall prevent conditions hazardous to personnel and detrimental to the equipment. Adequate provisions shall be furnished to prevent corrosive liquid, air, foreign particles, or condensation from entering the indicating circuit of the instrument. When the operating temperature of steam or service liquid exceeds 250° Fahrenheit (F), condensation pots shall be provided where suitable. Screw threads shall conform to FED-STD-H28. Instrument pressure piping shall be in accordance with ANSI B31.1. Back pressure and straight pipe requirements for satisfactory meter operation shall be recommended by the manufacturer. The meters shall be designed in accordance with the fluid properties, system operating conditions, and pipe diameter specified (see 6.2). Primary sensing elements for type I meters shall conform to the requirements of ASME PTC 19.5.

3.4 Interchangeability. All meters of the same classification, furnished with similar options under a specific contract, shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and spare parts.

3.5 First article inspection. When specified (see 6.2), the contractor shall furnish one meter of each classification specified for first article inspection and approval (see 4.2.1 and 6.4).

3.6 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.6.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fastenings shall be of the same material as the metals jointed or shall be cathodic to the metals joined.

3.7 Construction. Materials of construction for parts subject to pressure shall be selected in accordance with the operating medium and the system operating temperature and pressure. Moving parts shall be properly alined and adequately lubricated. Interior parts shall be accessible for adjustment and replacement. Means for compensating variances in viscosity, pressure, and temperature shall be provided as required to meet the accuracy requirements of these measurements. The indicator shall include mounting brackets, or other means for mounting, and shall be mounted either remotely or at the measuring location as specified (see 6.2). Electrical connection and input characteristics shall be as specified (see 6.2). When specified (see 6.2), means shall be provided to protect the meter from overrange and reversal of flow. Perform-

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ance shall be such that the primary sensing element will be capable of covering the specified range without replacement. When mercury or other fluid is used in the transmission of the sensing signal, provisions for draining the mercury and means for sealing the transmitter shall be furnished. When electrical components are used, normal variations in voltage and frequency shall not cause malfunction of the meter. When air transmission is provided, the indicating circuit shall be adequately protected from surges.

3.8 Performance. The meter shall be of the types and styles listed in table I, and shall operate in accordance with the requirements specified herein.

3.9 System accuracy. Meter accuracy shall be as specified in table I. The accuracy of the type I meters shall be applicable over the entire scale range. The accuracy of the type II meters shall be based on the requirements of table I for the scale range between 10 percent to 100 percent of capacity, see 4.5.1.

3.10 Details of components. The meter shall consist of accessory equipment specified (see 6.2), any additional apparatus normally furnished with this type of meter, and the components specified below.

3.10.1 Primary sensing element. The sensing element for type I shall be capable of creating a pressure differential compatible to the flow requirements for which the element has been designed. The sensing element for the type II meter shall consist of a vane or other device submerged in the system stream. The force of the fluid stream shall cause the vane to react and give an accurate indication of the fluid velocity. Transmission of the metered action shall be accomplished with minimum friction, pressure drop, or lost motion. When specified (see 6.2), straightening vanes shall be provided.

3.10.2 Indicator. The indicator shall directly or indirectly translate the measurement resulting from the action of the primary element as specified (see 6.2). The indicator shall be adjustable; wall, pedestal, or panel mounted, as specified (see 6.2); and shall clearly indicate the measured characteristic. Operation shall be accomplished either mechanically, hydraulically, pneumatically, or electrically. After being idle, the indicator shall be capable of continuous operation under a condition of 80 percent, or less, relative humidity. Wearing surfaces shall be so designed as to cause minimum friction, lost motion, or pressure drop. Normal system variations or slight distortion of the outer casing shall not cause the instrument to fail to indi- cate within the accuracy limits specified herein. Unless otherwise specified (see 6.2), each indicator shall be calibrated in the manufacturer's standard method.

3.10.3 Transmitting apparatus. Unless otherwise specified (see 6.2), operation of the transmitter shall be by the same means as employed for the indicator. The operating parts of the apparatus shall be constructed of non-corroding material and shall be enclosed in a metal casing. The transmitter shall convert the primary signal into units appropriate for the specified indication and shall not be adversely affected by slight distortion of the casing or normal system variations. When an electrical or pneumatic transmitter is specified, the electrical or air characteristics and connections shall be as specified (see 6.2). The transmitting device shall be arranged to avoid undue strain on the indicating apparatus.

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3.10.4 Recorder. When specified (see 6.2), an adjustable recorder, pneumatically or electrically operated, shall be provided as a combined indicating-recording unit for remote metering of the system fluid. Unless otherwise specified (see 6.2), the recorder shall be enclosed in a dustproofed casing and shall be equipped with a hinged, sealed door, capable of being secured, and shall have a nonshattering, high impact viewing window. The recording mechanism assembly shall be complete and include as main elements a removable marking pen, ink, lifter, charts, and an electrical or mechanical drive arrangement. The marker mechanism shall be corrosion-resistant steel and shall be easily accessible for maintenance. Unless otherwise specified (see 6.2), the chart shall be easily replaced and make one complete revolution in 24 hours. The chart scale shall be identifiable and graduated in units ranging from 0 to 100 percent of the specified maximum measurement. When specified for steam systems (see 6.2), the scale shall be graduated in pounds per hour (pph), with steam flow indicated in red, and airflow indicated in blue. The quantity and size of charts shall be as specified (see 6.2).

3.10.5 Flow computer. When specified, (see 6.2), a flow computer shall be provided that will furnish continuous and instantaneous analog or digital read-out for the differential head flow equations for orifice, flow nozzle, venturi, or turbine sensing element.

3.10.6 Integrating mechanism. When specified (see 6.2), an adjustable, round or straight reading totalizer shall be provided to accurately and clearly indicate the accumulated measurement recorded. A slight distortion of the casing shall not affect the operation of the mechanism. The mechanism shall be resettable. The internal mechanism shall be of corrosion-resistant materials. Gears, pinions, and segments shall be designed for smooth, positive mesh. The totalizer shall be enclosed in a housing designed to prevent the accumulation of moisture or fog on the dial glass.

3.10.7 Piping. Unless otherwise specified (see 6.2), the meter shall be furnished complete with necessary integral piping for installation into the required pipeline. When the system piping is tapped for instrument transmission piping, the instrument piping shall be pitched slightly to allow for adequate drainage way from the instrument. Piping for accessory pressure and temperature gages shall be as specified (see 6.2).

3.10.8 Bearings. Bearings provided in any of the meter components shall be designed for long life and resistance to wear. Bearing selection shall be based on the accuracy of fit required, ease of replacement, and ability to maintain true alinement.

3.11 Treatment and painting Unless otherwise specified (see 6.2), the meter shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the meter other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.12 Hydraulic and pneumatic equipment. Each system shall be furnished complete with necessary integral valving and piping for normal operation of the meter. Filtering of the operating medium for purposes of meter protection shall be in accordance with the manufacturer's standard practice.

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3.13 Gearing. When helical, herringbone, bevel, or worm gearing is used, the dimensions and tolerances shall conform to AGMA 390.03. Gears shall be firmly mounted, fully meshed, and adequately balanced to provide vibration-free operation.

3.14 Frost protection. When specified (see 6.2), meters shall be furnished with a frost protection device that is designed to yield under normal freezing conditions and reduce damage to meter parts.

3.15 Fungus control. When specified (see 6.2), fungus-proofing in accordance with MIL-T-152 shall be required.

3.16 Tamper-proofing. When specified (see 6.2), provisions shall be made for sealing the meter against tampering by unauthorized personnel.

3.17 Identification marking. Identification shall be permanently and legibly marked directly on the meter or on a corrosion-resisting metal plate securely attached to the meter at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.18 Workmanship.

3.18.1 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.18.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.19 Surface finish. Surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and made free of sand, dirt, fins, sprues, scale, flux, or other extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled. Measuring surfaces shall be finely ground, consistent with the intended meter accuracy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one meter when a first article is required (see 3.5 and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All meters offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for tests. Tests shall be based on inspection level S-2 and an Acceptable Quality Level (AQL) of 1.5 percent defective.

4.3.2 Sampling for examination. Examination shall be based on inspection level II and an AQL of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Examination. Each sample selected shall be examined for defects listed in table II.

TABLE II. Classification of defects.

Categories	Defects	Requirement paragraph
Major:		
101	Meter shall not be affected by vibration	3.1
102	Like parts not interchangeable	3.4
103	Materials not as specified	3.6
104	Dissimiliar metals likely to cause deterioration of parts	3.6.1

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TABLE II. Classification of defects - Continued.

Categories	Defects	Requirement paragraph
Major:		
105	Parts subject to system operating temperatures and pressure. Proper alinement and lubrication of moving parts	3.7
106	Failure to obtain an accurate indication	3.9
107	Sensing element not designed in accordance with system requirements	3.10.1
108	System variations or distortion of outer casing shall not cause instrument to fail to indicate within accuracy limits specified	3.10.2
109	Means of operation for transmitter to be the same as employed for indicator	3.10.3
110	When electrical or pneumatic transmitter specified, electrical or air characteristics connections shall be as specified	3.10.3
111	Recorder not functioning properly	3.10.4
112	Meter shall be furnished complete with integral piping	3.10.7
113	Hydraulic and pneumatic piping not complete	3.12
114	Mounting, meshing, and balancing of gears	3.13
115	Frost protection device to reduce damage to meter parts	3.14
116	Castings to be sound and free from defects	3.18.1
117	Boltholes accurately punched or drilled	3.18.2
Minor:		
202	Surface of castings, forgings, molded parts, stampings, and welded parts to be cleaned and made free of sand, dirt, fins, sprues, scale, flux, or other extraneous material	3.19

4.5 Tests. Each meter selected shall be tested and any unit failing to pass the following tests, as applicable, shall be rejected.

4.5.1 Accuracy. The meter shall be tested for accuracy of registration of the high and the low limits of the scale range as specified in 3.9. The meter shall be calibrated prior to the test and checked after its completion to determine the ability of the meter to maintain the calibrated setting. The test fluid characteristics, including test operating pressure and temperature conditions, shall be established on the basis of the specified system requirements. Flow conditions shall be simulated by the vendor, utilizing his manufacturing standards, when test facilities are not available to check actual system flow. The uninterrupted running time of the tested meter shall be not less than 72 hours with complete scanning of the scale range for each 24-hour period.

4.5.2 Component test. Instrument piping shall be tested in accordance with ANSI B31.1. Electrical component testing shall be based on the requirements of ANSI C39.4.

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4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each meter shall be packaged in a close-fitting box conforming to PPP-B-636, class-weather-resistant. Joints and seams of the boxes shall be waterproof sealed with tape in accordance with the appendix of the box specification.

5.1.2 Commercial. Material shall be packaged in accordance with MIL-STD-1188.

5.2 Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-794.

5.2.2 Commercial. Material shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I meters. Type I meters are primarily used for indicating, recording, and totalizing steam pressure, temperature and flow, airflow, feedwater flow, and demineralizer flow. Steam conditions ranging from 7,000 pph at 160 pounds per square inch (psi) to 135,000 pph at 600 psi and 800°F, and feedwater flow up to 1,000 gallons per minute are typical steam plant system conditions. Gas flow up to 88,000 cubic feet per hour at 250 psi is a typical gas system condition.

6.1.2 Type II meters. Type II meters are used for liquid service, such as in fuel oil, gasoline, and feedwater systems. Liquids under high pressures within the limits specified, whether viscous or corrosive, are covered by this type of meter.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and style meter required (see 1.2).
c. Specifications and standards (see 2.1.1).
d. Component configuration required to measure the systems fluid properties (see 3.1).
e. Fluid properties, system operating conditions, and pipe diameter required (see 3.3).
f. When first article is required for inspection and approval (see 3.5).
g. Whether indicator shall be mounted locally or remotely, electrical connections and input characteristics required, and whether protection from overrange and flow reversal are required (see 3.7).

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h. Accessory equipment required (see 3.10).
i. Whether straightening vanes are required (see 3.10.1).
j. Whether indication is direct or indirect, type of mounting for indicator, and whether indicator shall be calibrated other than as specified (see 3.10.2).
k. Whether operation of the transmitter is other than as specified, and electrical or air characteristics and connections required for the transmitter (see 3.10.3).
l. When a recorder is required, and type of case, if other than specified; and chart replacement, time for revolution, graduations, quantities, and sizes required (see 3.10.4).
m. Whether a flow computer is required (see 3.10.5).
n. Type of integrating mechanism, if required (see 3.10.6).
o. Meter piping and accessory gage piping required (see 3.10.7).
p. Treatment and painting, if other than specified (see 3.11).
q. Whether frost protection is required (see 3.14).
r. Whether fungus-proofing is required (see 3.15).
s. Whether meter is to be tamper-proof (see 3.16).
t. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Options. Overall accuracy of the meters specified herein are governed by the type of sensing element, the method of transmission, and the location of the indicating or recording element. A complete metering unit should, therefore, be obtained for a particular system application. Orifices and flow nozzles are normally used to meter high-velocity fluids. Venturi-type elements, which have low pressure-drop characteristics, may be used to meter fluids containing solids in suspension. The turbine- and propeller-type elements are most effective in applications where the flow of the liquid is of sufficient magnitude to make the pressure loss insignificant in relation to the overall system pressure.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of a type I or type II meter as described in table I. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-18221C(YD)	METERS, FLOW RATE INDICATING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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